

ORIGINAL RESEARCH

Intravenous Nicorandil in Patients With ST-Segment Elevation Myocardial Infarction Undergoing Primary PCI

The CLEAN Randomized Clinical Trial

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ABSTRACT

BACKGROUND Nicorandil, an adenosine triphosphate-sensitive potassium-channel opener with nitrate-like properties, may reduce reperfusion injury and microvascular obstruction in ST-segment elevation myocardial infarction (STEMI), but large-scale randomized evidence on long-term clinical outcomes is inconclusive.

OBJECTIVES The CLEAN trial aimed to assess whether adjunctive intravenous nicorandil improves 12-month clinical outcomes in patients with STEMI undergoing primary percutaneous coronary intervention.

METHODS In this multicenter, randomized, double-blind, placebo-controlled trial conducted at 49 hospitals in China, patients aged 18 to 80 years with STEMI within 12 hours of symptom onset were randomly assigned (1:1) to receive intravenous nicorandil (6 mg bolus before reperfusion followed by 6 mg/h infusion for 48 h) or matching placebo. Oral nicorandil was prohibited during follow-up. The primary outcome was a composite of cardiovascular death, nonfatal myocardial infarction, target vessel revascularization, or unplanned hospitalization for heart failure within 12 months.

RESULTS Between January 2021 and December 2023, 1,503 patients were enrolled and randomly assigned to nicorandil (n = 748) or placebo (n = 755). The primary composite outcome occurred in 98 patients (13.1%) in the nicorandil group (113 events over 717.2 person-years) and 99 (13.1%) in the placebo group (136 events over 710.3 person-years), with no significant difference between groups (rate ratio: 0.869; 95% CI: 0.650-1.162; *P* = 0.3429). Among secondary outcomes, nominal reductions were observed in cardiovascular death (1.9% vs 3.6%; HR: 0.515; 95% CI: 0.269-0.983) and target-vessel revascularization (1.1% vs 3.0%; HR: 0.322; 95% CI: 0.143-0.727), whereas rates of nonfatal myocardial infarction and unplanned hospitalization for heart failure were similar between groups. Adverse events did not differ between groups.

CONCLUSIONS In patients with STEMI undergoing primary percutaneous coronary intervention, adjunctive intravenous nicorandil did not significantly reduce the 12-month primary composite outcome. These findings do not support routine use of intravenous nicorandil in unselected patients with STEMI. (Clinical Efficacy and sAfeTy of Intravenous Nicorandil; [NCT04665648](https://doi.org/10.1016/j.jacc.2026.07.005)) (JACC. 2026;■:■-■) © 2026 by the American College of Cardiology Foundation.

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**ABBREVIATIONS
AND ACRONYMS****CMR** = cardiac magnetic resonance**CMVO** = coronary microvascular obstruction**HF** = heart failure**K-ATP** = adenosine triphosphate-sensitive potassium**LAD** = left anterior descending artery**PCI** = percutaneous coronary intervention**RR** = rate ratio**STEMI** = ST-segment elevation myocardial infarction**TIMI** = Thrombolysis in Myocardial Infarction**TVR** = target vessel revascularization

Primary percutaneous coronary intervention (PCI) is the preferred reperfusion strategy for patients with ST-segment elevation myocardial infarction (STEMI).¹ Despite restoration of epicardial blood flow, myocardial reperfusion at the microvascular level is often incomplete—this is a phenomenon known as no-reflow, primarily driven by coronary microvascular obstruction (CMVO). No-reflow can also result from distal embolization, vasospasm, or iatrogenic dissection, but CMVO is the central pathophysiological mechanism. Both CMVO and no-reflow are associated with larger infarct size, adverse left ventricular remodeling, and worse clinical outcomes, including heart failure (HF) and death.^{2,3} To date, no therapy has been definitively proven to prevent or effectively treat CMVO in clinical practice.⁴

Nicorandil, a hybrid adenosine triphosphate-sensitive potassium (K-ATP) channel opener and nitrate, has potential cardioprotective effects.⁵ Its nitrate moiety induces vasodilation of epicardial coronary arteries and reduces cardiac preload and afterload, while its K-ATP channel-opening property mimics ischemic preconditioning, potentially mitigating reperfusion injury at the mitochondrial and cellular level.^{6,7} These properties may attenuate CMVO and improve myocardial reperfusion. Previous small-to-medium-sized randomized trials suggested that intravenous nicorandil might improve coronary flow, promote ST-segment resolution, and preserve left ventricular function in patients with STEMI undergoing primary PCI.⁸⁻¹⁰ However, the J-WIND (Human atrial natriuretic peptide and nicorandil as adjuncts to reperfusion treatment for acute myocardial infarction) trial—the largest previous study—found that intravenous nicorandil did not reduce infarct size or improve clinical outcomes after 2.5 years of follow-up.¹¹ Several factors might explain these discrepant findings, including differences in nicorandil dosage, duration of infusion, and trial design. The J-WIND protocol used a lower-dose, had a shorter-duration regimen, and permitted subsequent oral nicorandil therapy, potentially confounding the assessment of the intravenous intervention.

The CLEAN (Clinical Efficacy and sAFety of intravenous Nicorandil) trial was designed to determine whether a high-dose, prolonged intravenous nicorandil regimen administered during primary PCI improves long-term clinical outcomes in patients with STEMI.

METHODS

STUDY DESIGN AND PARTICIPANTS. The CLEAN trial was an investigator-initiated, multicenter, randomized, double-blind, placebo-controlled study conducted at 49 tertiary hospitals in China. The trial design and rationale have been published previously,¹² and a schematic diagram is provided in [Supplemental Figure 1](#). A complete list of participating centers and trial personnel is available in the Appendix. The trial was registered at ClinicalTrials.gov (NCT04665648) and complied with the principles of the Declaration of Helsinki and Good Clinical Practice guidelines. The study protocol was approved by the ethics committee at each participating center, including the Zhongshan Hospital Ethics Committee (Approval No. B2020-270R) and all other site committees (full list in the [Supplemental Appendix](#)).

Eligible patients were adults aged 18 to 80 years with a diagnosis of STEMI within 12 hours of symptom onset who were scheduled for primary PCI. STEMI was defined as persistent ST-segment elevation of at least 0.1 mV in 2 or more contiguous leads or a new left bundle-branch block, accompanied by symptoms of myocardial ischemia and/or elevated biomarkers of myocardial necrosis.

Key exclusion criteria included any contraindication to nicorandil (hypersensitivity, hypotension [systolic blood pressure ≤ 100 mm Hg at admission], cardiogenic shock, or moderate-to-severe liver disease), use of nicorandil within the previous 6 months, contraindications to PCI, requirement for dialysis, pregnancy or lactation, and participation in another investigational drug or device study within 30 days. Full details of the inclusion and exclusion criteria are provided in the [Supplemental Appendix](#). All patients provided written informed consent before enrollment. In the emergency setting, verbal informed consent was obtained from the patient or a legal representative, witnessed by an independent

The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the [Author Center](#).

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physician, followed by written consent as soon as feasible after PCI.

RANDOMIZATION AND MASKING. Eligible patients were randomly assigned in a 1:1 ratio to receive either intravenous nicorandil or matching placebo. Randomization was performed using a centralized, web-based system with a concealed allocation sequence. The randomization sequence was computer-generated using permuted blocks of 4, stratified by study site to ensure balanced treatment assignment within each center.

Nicorandil and placebo were supplied as powders for injection by Sihuan Kebao Pharmaceutical (Beijing, China) and were identical in appearance. Study drugs were reconstituted with saline by clinical staff not involved in outcome assessment. Patients, investigators, emergency physicians, interventional cardiologists, outcome assessors, and data analysts were masked to treatment assignment throughout the trial.

PROCEDURES. Patients in the nicorandil group received an intravenous bolus of 6 mg nicorandil immediately before coronary reperfusion (balloon inflation or thrombus aspiration), followed by a continuous intravenous infusion at 6 mg/h for 48 h after the index PCI procedure (unless intolerance, eg, hypotension). Patients in the placebo group received an identical volume of 0.9% saline as a bolus and continuous infusion. The dose and duration were selected based on previous pharmacokinetic and clinical studies suggesting optimal effects on coronary blood flow and cardioprotection with this regimen.⁹⁻¹⁴

All patients underwent primary PCI according to contemporary guideline recommendations. The choice of vascular access, anticoagulation regimen, stent type, and use of adjunctive devices (eg, thrombus aspiration, intra-aortic balloon pump) was at the discretion of the operating interventional cardiologist. All patients received guideline-directed medical therapy, including a loading dose of aspirin (300 mg) and a P2Y₁₂ inhibitor (ticagrelor 180 mg or clopidogrel 600 mg) before PCI, followed by maintenance dual antiplatelet therapy. Use of statins, beta-blockers, angiotensin-converting enzyme inhibitors, or angiotensin receptor blockers was strongly recommended unless contraindicated. Oral nicorandil was prohibited during the 12-month follow-up period.

OUTCOMES. The primary efficacy outcome was a composite of major adverse cardiovascular events, defined as cardiovascular death, nonfatal myocardial infarction, target vessel revascularization (TVR), or

unplanned hospitalization for HF within 12 months after randomization.

Secondary efficacy outcomes were as follows: 1) the individual components of the primary composite outcome; 2) incidence of slow coronary flow after PCI, defined as a corrected Thrombolysis in Myocardial Infarction (TIMI) frame count >20 frames in the culprit artery in the absence of mechanical obstruction; and 3) complete ST-segment resolution measured 2 h after PCI. ST-segment resolution was assessed as an ordinal variable: <50% (none), 50% to 70% (partial), and ≥70% (complete). Complete resolution was defined as ≥70% reduction in summed ST-segment elevation compared with the pre-PCI electrocardiogram. A sensitivity analysis using a binary threshold of ≥50% was also performed.

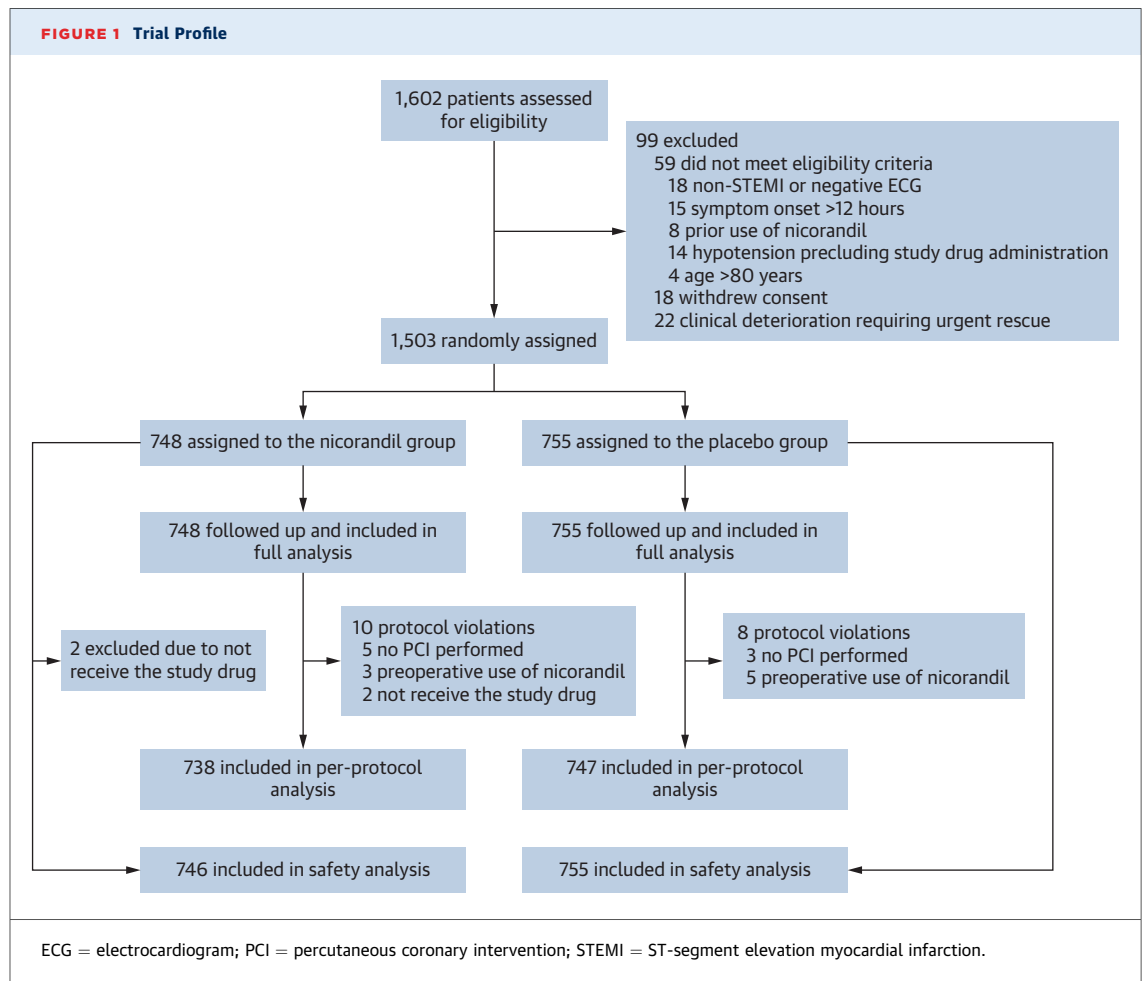
Safety outcomes included the incidence of hypotension (systolic blood pressure <90 mm Hg or a decrease >30 mm Hg from baseline requiring intervention) during or after drug infusion, and the occurrence of other adverse events.

All suspected primary and secondary outcome events were adjudicated by an independent clinical events committee masked to treatment assignment, using prespecified definitions in the Appendix aligned with the 2017 Cardiovascular and Stroke Endpoint Definitions for Clinical Trials¹⁵ and the Fourth Universal Definition of Myocardial Infarction.¹⁶

STATISTICAL ANALYSIS. The sample size was calculated based on simulations for the proportional rates model. Assuming an annualized rate of 14.6% for recurrent events and 3% for cardiovascular death in the control group, based on previous nicorandil trials,^{9,11} we targeted a 27% reduction in the rate ratio (RR) for the primary composite endpoint (corresponding approximately to a 30% reduction in recurrent events and a 10% reduction in cardiovascular death). A total of 1,500 patients was estimated to provide 80% power at a 2-sided alpha level of 0.05, allowing for 10% random censoring.

Primary and secondary outcome analyses were performed in the full analysis set, which included all randomly assigned patients according to the intention-to-treat principle. Safety analyses were performed in the safety population, comprising all patients who received at least 1 dose of study drug or placebo. Categorical variables are presented as numbers (%) and continuous variables as mean ± SD or median (IQR), as appropriate.

The primary outcome was analyzed using the semiparametric proportional rates model (Lin-Wei-Yang-Ying method) without adjustment for site. In the Lin-Wei-Yang-Ying model, cardiovascular



death was treated as an event within the recurrent-event process and noncardiovascular death was treated as censoring. Ghosh-Lin cumulative incidence curves were generated for graphical presentation of recurrent events. Three post-hoc sensitivity analyses were conducted to verify the stability of the primary endpoint by using different models. With 2 prespecified interim analyses using Haybittle-Peto stopping boundaries, the 2-sided significance level for the primary outcome was set at $P < 0.0497$.

Time-to-first-event analyses for the primary endpoint and its components were performed using Cox proportional hazards models, adjusted for site. Recurrent event analyses were conducted using negative binomial regression. The incidences of slow coronary flow and complete ST-segment resolution were compared using binary or ordinal logistic regression.

Prespecified subgroup analyses assessed treatment effect on the primary outcome according to baseline characteristics in the [Supplemental Appendix](#). No adjustment for multiple testing was made for

subgroup or secondary outcome analyses; these results should be interpreted as exploratory.

An independent data and safety monitoring board oversaw the trial, and an independent clinical events committee adjudicated all clinical events in a blinded manner in the [Supplemental Appendix](#). All statistical analyses were performed using SAS version 9.4 (SAS Institute, Inc) and R version 4.4.1 (R Foundation for Statistical Computing).

ROLE OF THE FUNDING SOURCE. The funders of the study had no role in study design, data collection, data analysis, data interpretation, or writing of the report. The corresponding authors had full access to all study data and had final responsibility for the decision to submit for publication.

RESULTS

Between January 15, 2021, and December, 10, 2023, 1,602 patients were assessed for eligibility, of whom 1,503 were enrolled and randomly assigned: 748 to the nicorandil group and 755 to the placebo group

(Figure 1). All randomized patients were included in the full analysis set for efficacy analyses. The safety population comprised 746 patients in the nicorandil group and 755 in the placebo group; 2 patients in the nicorandil group did not receive the study drug because of consent withdrawal after randomization.

Adherence to the 48-h infusion protocol was high and similar between groups. The mean proportion of planned infusion volume received was 93.6% in the placebo group and 91.2% in the nicorandil group. Premature permanent discontinuation of the study drug due to an adverse event occurred in 13 placebo patients (1.7%) and 20 nicorandil patients (2.7%). All patients were included in the intention-to-treat analysis regardless of adherence.

Baseline demographic and clinical characteristics were well balanced between the 2 groups (Table 1). The median age of the study population was 58.4 years (IQR: 51.0-67.0 years); 1,268 (84.4%) patients were male and 235 (15.6%) were female. Diabetes mellitus was present in 467 patients (31.1%), with similar proportions in the placebo (228 [30.2%]) and nicorandil (239 [32.0%]) groups. The median time from symptom onset to hospital arrival was 2.87 h (IQR: 1.47-4.92 h). Anterior wall myocardial infarction was the most common presentation, occurring in 809 patients (53.8%). Mean systolic and diastolic blood pressure at admission were 138.2 mm Hg and 86.1 mm Hg, respectively.

Procedural characteristics were similar between treatment groups (Table 2). The left anterior descending artery was the most frequent culprit vessel, involved in 779 patients (51.8%). Preprocedural TIMI flow grade 0 or 1 was present in 1,093 patients (72.7%). Thrombus aspiration was performed in 304 patients (20.2%), and stent implantation in 1,298 (86.4%). Postprocedural TIMI flow grade 3 was achieved in 1,433 patients (95.3%), with no difference between groups.

The primary composite endpoint—cardiovascular death, nonfatal myocardial infarction, TVR, or unplanned hospitalization for HF within 12 months—occurred in 98 patients (13.1%) in the nicorandil group and 99 patients (13.1%) in the placebo group, comprising 113 events over 717.2 person-years of follow-up and 136 events over 710.3 person-years of follow-up, respectively. There was no significant difference between the 2 groups (RR: 0.869; 95% CI: 0.650-1.162; $P = 0.3429$) (Table 3). Ghosh-Lin mean cumulative event curves for the primary outcome are shown in Figure 2. Time to first occurrence of the primary composite endpoint also did not differ significantly between groups (HR: 0.998; 95% CI: 0.755-1.320) (Supplemental Figure 2). Prespecified

TABLE 1 Baseline Characteristics of the Study Population

	Nicorandil (n = 748)	Placebo (n = 755)	Total (N = 1,503)
Demographics			
Age, y	58.2 ± 12.1	58.6 ± 11.8	58.4 ± 11.9
Male	641 (85.7)	627 (83.0)	1268 (84.4)
Body mass index, kg/m ²	25.1 ± 3.5	25.0 ± 3.6	25.0 ± 3.5
Medical history			
Hypertension	420 (56.1)	416 (55.1)	836 (55.6)
Diabetes mellitus	239 (32.0)	228 (30.2)	467 (31.1)
Dyslipidemia	140 (18.7)	172 (22.8)	312 (20.8)
Current smoker	379 (50.7)	385 (51.0)	764 (50.8)
Previous myocardial infarction	25 (3.3)	18 (2.4)	43 (2.9)
Previous PCI	37 (4.9)	31 (4.1)	68 (4.5)
Clinical presentation			
Systolic blood pressure, mm Hg	138.2 ± 22.8	138.2 ± 22.2	138.2 ± 22.5
Diastolic blood pressure, mm Hg	86.0 ± 16.2	86.2 ± 15.0	86.1 ± 15.6
Heart rate, beats/min	80.5 ± 16.5	79.3 ± 16.5	79.9 ± 16.5
Infarct location			
Anterior	425 (56.8)	384 (50.9)	809 (53.8)
Inferior	296 (39.6)	340 (45.0)	636 (42.3)
Other	27 (3.6)	31 (4.1)	58 (3.9)
Laboratory values			
White blood cell count, ×10 ⁹ /L	10.4 ± 3.5	10.5 ± 3.3	10.5 ± 3.4
Hemoglobin, g/L	148.2 ± 16.8	148.1 ± 18.6	148.1 ± 17.7
Creatinine, μmol/L	78.6 ± 30.5	78.7 ± 37.2	78.6 ± 34.1
Time to presentation			
Symptom onset to hospital arrival, h	2.68 (1.50-4.73)	2.98 (1.45-5.00)	2.87 (1.47-4.92)
Values are mean ± SD, n (%), or median (IQR). PCI = percutaneous coronary intervention.			

TABLE 2 Procedural Characteristics and Medications

	Nicorandil (n = 748)	Placebo (n = 755)	Total (N = 1,503)
Infarct-related artery			
Left anterior descending	412 (55.1)	367 (48.6)	779 (51.8)
Left circumflex	75 (10.0)	87 (11.5)	162 (10.8)
Right coronary	238 (31.8)	282 (37.4)	520 (34.6)
Pre-PCI TIMI flow grade			
Grade 0	487 (65.1)	506 (67.0)	993 (66.1)
Grade 1	54 (7.2)	46 (6.1)	100 (6.7)
Grade 2	64 (8.6)	69 (9.1)	133 (8.8)
Grade 3	123 (16.4)	118 (15.6)	241 (16.0)
Procedural details			
Thrombus aspiration	155 (20.7)	149 (19.7)	304 (20.2)
Stent implantation	649 (86.8)	649 (86.0)	1,298 (86.4)
Post-PCI TIMI flow grade 3	713 (95.3)	720 (95.4)	1,433 (95.3)
Key medications at index procedure			
Aspirin	739 (98.8)	736 (97.5)	1,475 (98.1)
P2Y ₁₂ inhibitor			
Ticagrelor	614 (82.1)	626 (82.9)	1,240 (82.5)
Clopidogrel	134 (17.9)	129 (17.1)	263 (17.5)
Parenteral anticoagulant			
Heparin	604 (80.7)	604 (80.0)	1,208 (80.3)
Bivalirudin	50 (6.7)	51 (6.8)	101 (6.7)
Values are n (%). TIMI = Thrombolysis in Myocardial Infarction; other abbreviations as in Table 1.			

TABLE 3 Primary and Secondary Efficacy Outcomes

Outcome	Nicorandil (n = 748)	Placebo (n = 755)	RR (95% CI)	HR (95% CI) ^b	OR (95% CI)	Absolute Risk Difference (95% CI)	P Value ^a
Primary outcome							
Cardiovascular death, nonfatal MI, TVR, or unplanned HF hospitalization within 12 mo	98 (13.1)/113 (717.2 PY)	99 (13.1)/136 (710.3 PY)	0.869 (0.650-1.162)	-	-	-	0.3429
Secondary outcomes							
Individual components							
Cardiovascular death	14 (1.9)	27 (3.6)	-	0.515 (0.269-0.983)	-	-1.7% (-3.45% to -0.04%)	0.0443
Nonfatal MI	10 (1.3)	9 (1.2)	-	1.148 (0.465-2.832)	-	0.1% (-1.08% to 1.39%)	0.7651
TVR	8 (1.1)	23 (3.0)	-	0.322 (0.143-0.727)	-	-2.0% (-3.55% to -0.54%)	0.0064
Unplanned HF hospitalization	74 (9.9)	66 (8.7)	-	1.149 (0.824-1.601)	-	1.2% (-1.80% to 4.12%)	0.4140
Recurrent events^c							
Nonfatal MI (recurrent)	11	9	1.211 (0.502-2.923)	-	-	-	0.6699
TVR (recurrent)	8	23	0.345 (0.154-0.771)	-	-	-	0.0095
Unplanned HF hospitalization (recurrent)	80	77	1.013 (0.716-1.433)	-	-	-	0.9410
Procedural outcomes							
Slow flow/no-reflow post-PCI	77 (10.3)	69 (9.1)	-	-	1.160 (0.815-1.650)	-	0.4097
Complete ST-segment resolution	412/727 (56.7)	429/727 (59.0)	-	-	0.920 (0.749-1.130)	-	0.4381
Exploratory outcomes							
All-cause death	19 (2.5)	32 (4.2)	-	0.589 (0.334-1.041)	-	-	0.0687

Values are n (%) / total number of events (total follow-up person-years), n (%), total number of events, or n/N (%), unless otherwise indicated. All analyses were performed in the full analysis set (intention-to-treat). ^aP values for secondary and exploratory outcomes are nominal and have not been adjusted for multiple comparisons. Therefore, these values should be interpreted as hypothesis generating. ^bHRs were estimated using the Cox proportional hazards model adjusted for study site. ^cData are total numbers of events.

HF = heart failure; MI = myocardial infarction; PY = person-years; RR = rate ratio; TVR = target-vessel revascularization; other abbreviations as in [Table 1](#).

and post-hoc sensitivity analyses yielded consistent results ([Supplemental Table 1](#)).

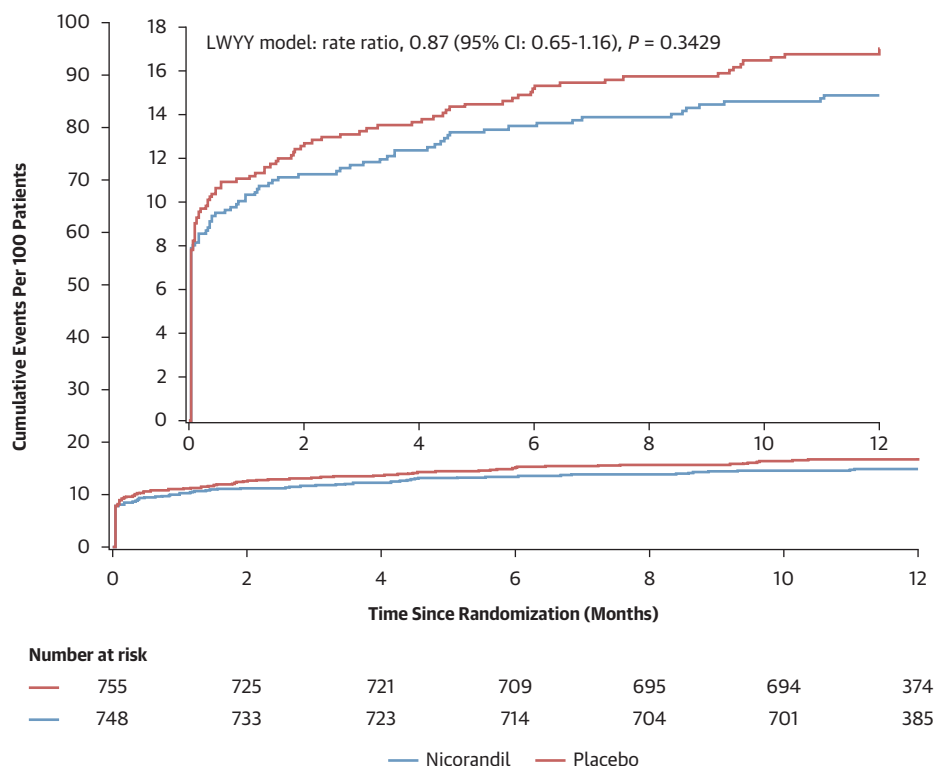
Secondary analyses of the individual components of the primary endpoint suggested differential patterns ([Table 3](#)). Kaplan-Meier curves for time to first event of each secondary outcome are shown in [Figure 3](#). Cardiovascular death occurred in 14 patients (1.9%) in the nicorandil group vs 27 patients (3.6%) in the placebo group (HR: 0.515; 95% CI: 0.269-0.983) ([Figure 3A](#)). TVR was also nominally lower in the nicorandil group (8 patients [1.1%] vs 23 patients [3.0%]; HR: 0.322; 95% CI: 0.143-0.727) ([Figure 3C](#)). No significant differences were observed for nonfatal myocardial infarction (10 patients [1.3%] vs 9 patients [1.2%]; HR: 1.148; 95% CI: 0.465-2.832) ([Figure 3B](#)) or unplanned hospitalization for HF (74 patients [9.9%] vs 66 patients [8.7%]; HR: 1.149; 95% CI: 0.824-1.601) ([Figure 3D](#)).

Recurrent event analyses yielded consistent findings ([Table 3](#)). The rate of TVR was lower in the nicorandil group (RR: 0.345; 95% CI: 0.154-0.771),

whereas rates of recurrent nonfatal myocardial infarction and unplanned hospitalization for HF did not differ between groups.

Procedural outcomes showed no statistically significant differences between treatment groups ([Table 3](#)). The incidence of slow flow or no-reflow phenomenon was similar (77 patients [10.3%] in the nicorandil group vs 69 patients [9.1%] in the placebo group; OR: 1.160; 95% CI: 0.815-1.650). Complete ST-segment ($\geq 70\%$) resolution at 2 h after PCI occurred in 412 patients (56.7%) receiving nicorandil and 429 patients (59.0%) receiving placebo (OR: 0.920; 95% CI: 0.749-1.130). A sensitivity analysis using a binary threshold of $\geq 50\%$ (partial or complete resolution) gave similar results ([Supplemental Table 2](#)).

All-cause mortality was numerically lower in the nicorandil group (19 patients [2.5%] vs 32 patients [4.2%]; HR: 0.589; 95% CI: 0.334-1.041) ([Table 3](#), [Supplemental Figure 3](#)). Biomarker and echocardiographic parameters are shown in [Supplemental Table 3](#). Peak creatine kinase-MB levels

FIGURE 2 Mean Cumulative Function of the Primary Composite Outcome

Ghosh-Lin estimated curves showing the mean cumulative number of recurrent events per subject for the primary composite outcome of cardiovascular death, nonfatal myocardial infarction, target-vessel revascularization, or unplanned hospitalization for heart failure within 12 mo. LWYY = Lin-Wei-Yang-Ying.

after PCI were similar between groups. N-terminal pro-B-type natriuretic peptide concentrations at 180 days, as well as left ventricular dimensions and ejection fraction, did not differ significantly between treatment groups.

Prespecified subgroup analyses showed no significant interactions between treatment assignment and any of the tested baseline characteristics, including age, sex, body mass index, diabetes, history of HF, previous myocardial infarction, infarct-related artery, time to reperfusion, pre-PCI TIMI flow, thrombus aspiration, or preticagrelor use (all interaction $P > 0.05$) (Figure 4, Supplemental Table 4). Although the treatment effect appeared numerically more pronounced in patients with diabetes (RR: 0.599; 95% CI: 0.362-0.992), the interaction test was not significant ($P = 0.0698$).

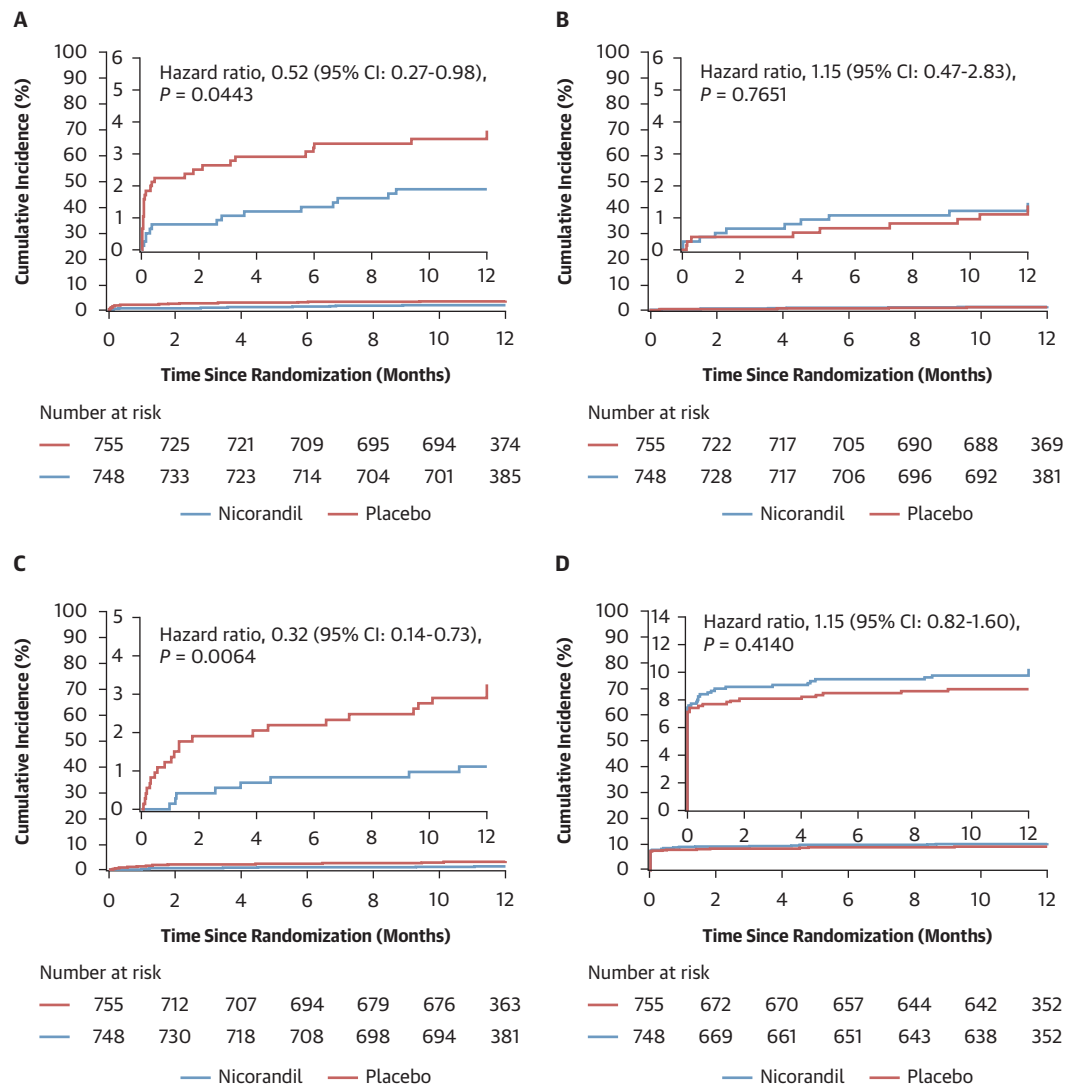
Safety outcomes are summarized in Table 4. The incidence of any treatment-emergent adverse event was 16.9% in the nicorandil group vs 13.8% in the placebo group ($P = 0.095$). Headache occurred more

frequently with nicorandil (8.2% vs 5.7%; $P = 0.052$). Rates of hypotension, serious adverse events, and treatment discontinuation due to adverse events were similar between groups.

DISCUSSION

In this large, multicenter, randomized, double-blind, placebo-controlled trial involving patients with STEMI undergoing primary PCI, adjunctive intravenous nicorandil did not significantly reduce the primary composite endpoint of cardiovascular death, nonfatal myocardial infarction, TVR, or unplanned hospitalization for HF at 12 months. These results are directly applicable to regions where nicorandil is already in use (Asia, Europe, UK, Australia) and argue against its introduction in North America without further evidence.

The primary outcome was neutral despite using a higher-dose, longer-duration regimen than the J-WIND trial, and despite prohibiting oral nicorandil

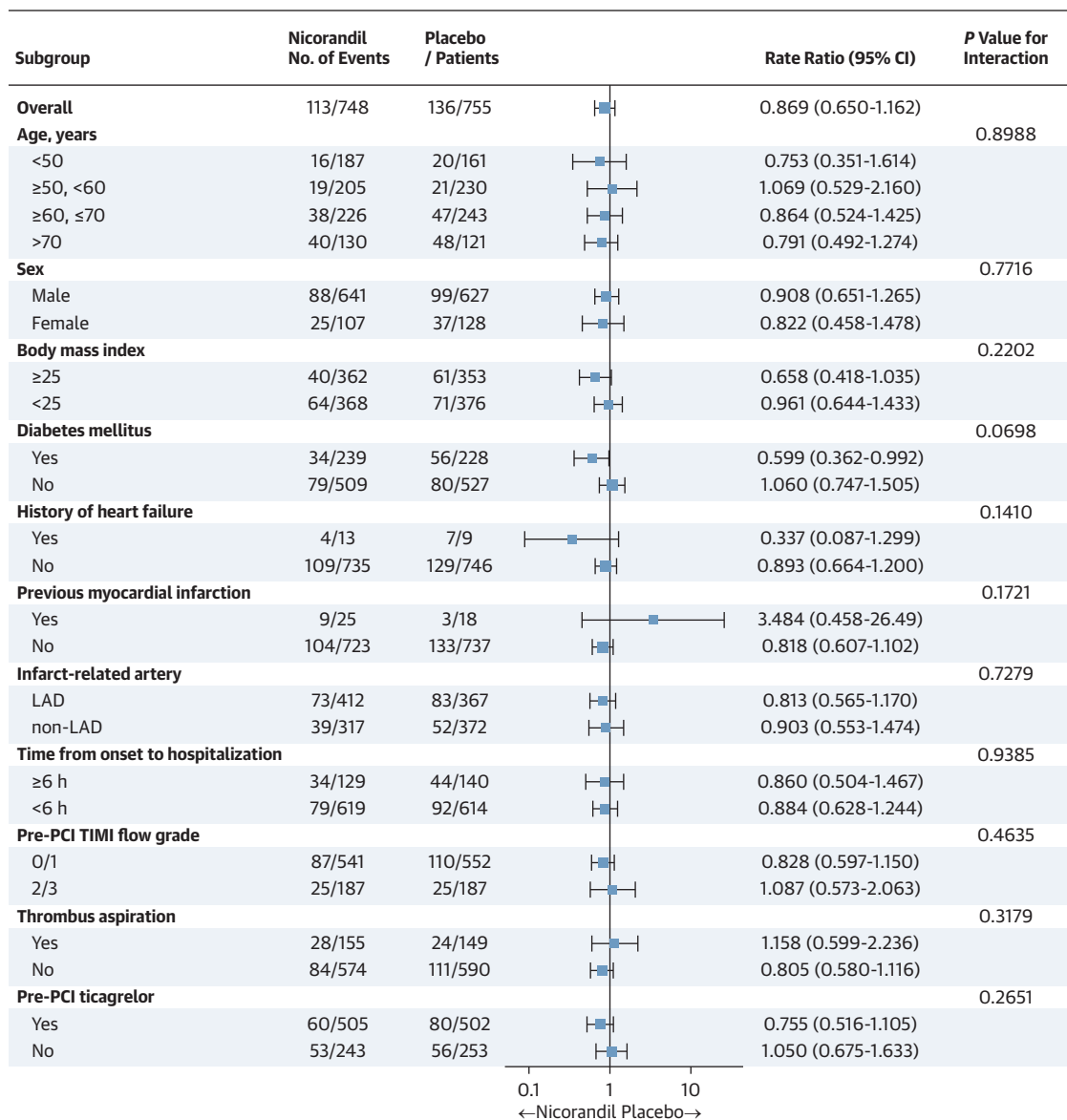
FIGURE 3 Kaplan-Meier Curves for the Secondary Outcome Components

Kaplan-Meier curves showing cumulative event rates for the individual components of the primary composite outcome: (A) cardiovascular death; (B) nonfatal myocardial infarction; (C) target-vessel revascularization; (D) unplanned hospitalization for heart failure. P values are nominal and not adjusted for multiple comparisons.

during follow-up. The absence of improvement in acute surrogate markers (slow flow, ST-segment resolution) and in HF hospitalization reinforces the robustness of the result. Exploratory analyses suggested nominally lower incidences of cardiovascular death and TVR with nicorandil, but these findings arise from small numbers of events and unadjusted comparisons. They should be considered hypothesis-generating and require independent validation. Potential mechanisms could include limitation of infarct size via K-ATP channel opening or reduced

inflammation, but the lack of effect on HF hospitalization and surrogate markers tempers enthusiasm.

Why was the primary outcome neutral despite a biologically plausible mechanism? Nicorandil combines nitrate-induced vasodilation with K-ATP channel opening, which in preclinical studies reduces reperfusion injury and limits infarct size. However, translation into clinical benefit has consistently failed in large-scale trials, not only for nicorandil but also for adenosine, nitroprusside, cyclosporine, and many other agents.¹⁷ Several factors explain this gap.

FIGURE 4 Prespecified Subgroup Analysis of the Primary Composite Outcome

LAD = left anterior descending artery; TIMI = Thrombolysis in Myocardial Infarction; other abbreviations as in [Figure 1](#).

First, the complexity of human CMVO extends beyond single-target pharmacology; it involves distal embolization, endothelial dysfunction, pericyte constriction, inflammation, and intramyocardial hemorrhage. Second, the timing and dose—even the intensified 48-h regimen used in CLEAN—may still be insufficient to modify the downstream cascade once reperfusion has occurred. Third, contemporary background therapy (rapid PCI, dual antiplatelet

therapy, statins, beta-blockers) has already raised the bar, making it difficult to demonstrate additive benefit.

Our neutral primary outcome aligns with the J-WIND trial, which used a lower-dose, shorter-duration nicorandil regimen (4 mg bolus + 4 mg/h for 24 h) and permitted oral nicorandil. CLEAN used a higher dose and longer duration without oral nicorandil, yet still no overall benefit was observed.

TABLE 4 Safety Outcomes

Outcome	Nicorandil (n = 746)	Placebo (n = 755)	P Value
Any TEAE	126 (16.9)	104 (13.8)	0.095
TEAE leading to permanent discontinuation	20 (2.7)	13 (1.7)	0.261
Hypotension	31 (4.2)	29 (3.8)	0.802
Headache	61 (8.2)	43 (5.7)	0.052
Serious adverse event	3 (0.4)	5 (0.7)	0.499
Drug-related serious adverse event	0	1 (0.1)	-

Values are n (%). Safety population includes all patients who received at least 1 dose of study drug or placebo.
TEAE = treatment-emergent adverse event.

Notably, the CHANGE (Effects of Nicorandil Administration on Infarct Size in Patients With ST-Segment-Elevation Myocardial Infarction Undergoing Primary Percutaneous Coronary Intervention) trial used the identical high-dose, prolonged intravenous nicorandil regimen (6 mg bolus + 6 mg/h for 48 h) as CLEAN and reported a significant reduction in infarct size measured using cardiac magnetic resonance (CMR), as well as a lower incidence of microvascular obstruction.¹⁰ However, CHANGE was not powered for clinical endpoints, and its positive findings on surrogate markers did not translate into a statistically significant reduction in the composite clinical outcome in CLEAN. This pattern mirrors other agents: adenosine showed improved surrogate markers in small studies (REOPEN-AMI [Intracoronary Nitroprusside Versus Adenosine in Acute Myocardial Infarction]) but failed to improve clinical outcomes in large trials (REFLO-STEMI [The REperfusion Facilitated by LOcal adjunctive therapy in STEMI]).^{18,19} Intracoronary nitroprusside, verapamil, and diltiazem have likewise not translated into hard endpoint benefits.^{2,20} Taken together, these observations suggest that improvement in CMR-derived infarct size or microvascular obstruction does not reliably predict a reduction in clinical events such as death, rehospitalization for HF, or revascularization in the contemporary era of rapid PCI and guideline-directed medical therapy.

Even high-risk subgroups did not derive benefit. Based on prior literature, we prespecified several subgroups of patients expected to have a higher likelihood of CMVO and therefore potentially greater benefit from nicorandil, including anterior myocardial infarction, prolonged ischemia, low pre-PCI TIMI flow (grade 0-1), and those who underwent thrombus aspiration.²¹ Across all these subgroups, the RRs for

the primary composite endpoint were not significantly different from the overall neutral result, and no significant treatment-by-subgroup interactions were observed. The nominal signal in patients with diabetes was not statistically significant after correction for multiple comparisons and should be considered hypothesis-generating. The absence of benefit even in these high-risk CMVO populations further underscores the robustness of the overall neutral conclusion and does not support a targeted strategy in any specific high-risk subgroup.

Although baseline characteristics were generally balanced, the nicorandil group had slightly higher proportions of anterior myocardial infarction (56.8% vs 50.9%) and left anterior descending artery (LAD) as the culprit artery (55.1% vs 48.6%). If anything, this imbalance would bias results against nicorandil, as anterior infarction is associated with worse outcomes. Despite this potential disadvantage, no benefit was observed, further reinforcing the neutral conclusion. Importantly, prespecified subgroup analyses based on infarct-related artery (LAD vs non-LAD) showed no significant treatment-by-subgroup interaction, indicating that this baseline imbalance did not materially affect the primary conclusion.

The trial population was predominantly male (84.4%), which aligns with the epidemiology of STEMI in China and other Asian populations, as consistently reported in large Chinese STEMI trials.^{22,23} This distribution does not reflect enrollment bias. The limited number of female participants, however, precludes meaningful sex-specific subgroup analyses.

What does CLEAN teach the field? This trial adds to a long list of failed cardioprotective strategies in STEMI. Several lessons emerge. First, preclinical models of reperfusion injury typically use young, healthy animals with no comorbidities, whereas real-world STEMI patients are older, have multiple risk factors, and present with complex pathophysiology. Second, the powerful background therapy in contemporary practice—including prompt primary PCI, dual antiplatelet therapy, high-intensity statins, and beta-blockers—may already optimize outcomes, leaving little incremental room for additional pharmacologic interventions. Third, the surrogate endpoints commonly used in early-phase trials (eg, ST-segment resolution, angiographic slow flow, CMR-derived infarct size) correlate only modestly with clinical outcomes. This discrepancy highlights the potential limitations of relying on imaging-based surrogates for hard clinical outcomes. Fourth, unselected patient populations may dilute potential

benefits; however, our subgroup analyses suggest that even in traditionally defined high-risk groups, benefit was not observed. Future studies should focus on alternative mechanisms or novel targets (eg, pericytes, neutrophil extracellular traps) and incorporate sensitive microvascular measures as both selection tools and endpoints.²⁰ By providing a definitive negative result across both unselected and high-risk populations, CLEAN redirects the field away from a “one-size-fits-all” periprocedural approach and toward more targeted, mechanistically guided investigation.^{24,25}

STUDY LIMITATIONS. The CLEAN trial was conducted during the COVID-19 pandemic; although this posed challenges to enrollment and on-site follow-up, data integrity and completeness were maintained through remote monitoring and close collaboration across sites, and the pandemic did not affect the validity of the trial results. Notwithstanding this successful mitigation of external challenges, several inherent limitations of the trial deserve consideration. First, the trial was conducted exclusively in China, which may limit generalizability to other populations. Second, we did not systematically perform CMR imaging or invasive coronary physiology assessments to directly link clinical outcomes with reductions in infarct size or microvascular obstruction. Third, we did not use a core laboratory for angiographic or electrocardiogram endpoints (slow flow, ST-segment resolution), introducing potential variability. Fourth, secondary and subgroup analyses were not adjusted for multiple comparisons; the nominal *P* values observed would not survive conservative correction and should be regarded as exploratory. Fifth, missing data for N-terminal pro-B-type natriuretic peptide and echocardiography (~20%-25%) were present, although missingness was balanced between groups. Sixth, the inclusion of TVR in the primary composite—a component not directly aligned with nicorandil’s proposed microvascular mechanism—may have contributed to the neutral result, but this was a prespecified choice to capture a comprehensive set of clinically relevant endpoints.

CLINICAL IMPLICATIONS. Routine adjunctive intravenous nicorandil does not improve 12-month clinical outcomes in unselected STEMI patients undergoing contemporary primary PCI and should

not be adopted as standard care. Future research should explore novel therapeutic targets beyond vasodilation and incorporate sensitive microvascular assessment tools.

CONCLUSIONS

In patients with STEMI undergoing primary PCI, adjunctive intravenous nicorandil did not significantly reduce the 12-month primary composite endpoint of cardiovascular death, nonfatal myocardial infarction, TVR, or unplanned hospitalization for HF. These findings do not support the routine use of intravenous nicorandil in unselected patients with STEMI, nor do they identify a high-risk subgroup that derives benefit.

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APPENDIX For supplemental tables and figures, please see the online version of this paper.